

Effects of Island Location on Adelie Penguin Size

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STA 160

Penguin Dataset Project

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Abstract

This report examines the Palmer Penguin Data Set, collected by Dr. Kristen Gorman in collaboration with the Palmer Station in Antarctica. I am focusing on three islands in the Palmer Archipelago: Bisco, Dream, and Tortgersen, and their possible effects on four body measurements: bill length, bill depth, body mass, and flipper length. Adelei penguins are the only species found on all three islands, making it a good way to analyze the species to see island effects on body measurements. Additionally, Torgeersen is the only island with only Adelei penguins. My hypothesis was that island location would have a significant effect on Adelie penguin body measurements, most likely stemming from environmental adaptation. Using methods such as one-way ANOVA, two-way ANOVA, histograms, ECDF, 3D plots, scatter plots, and boxplots, I found no significant effect of island location on penguin body measurements. However, using the same methods, I found that sex was a significant factor across all four body measurements, showing that sexual dimorphism is the main driver in Adelie penguin size.

Introduction

From initial research into the dataset, I learned that the Adelie penguins were the only species found on all three islands. This made the Adelie penguin species a perfect candidate to be compared across all islands to see the effects of the island on body size.

In biology, animals living in different geographic locations and conditions often develop different morphological differences due to environmental conditions like food availability, predator pressure, and climate. Charles Darwin noticed this in the Galapagos Islands with 17 species of finches on multiple islands, with morphological differences; some birds have larger wings, or shorter beaks. If the three islands, Torgersen, Biscoe, and Dream, have meaningful environmental differences, the Adelie penguins may have differing body sizes based on the island they live on. However, if the islands are ecologically similar, there may be no difference in body sizes on each island.

Another biological factor I examined was sexual dimorphism in Adelie penguins. If sexual dimorphism is present in Adelie penguins, it may explain more about body measurements than island locations.

Overall, my hypothesis is that island location has a significant effect on Adelie penguin body measurements.

Methods

My first step in my project was removing any NA values and filtering out the other two penguin species: Chinstrap and Gentoo. This left me with a dataset of 146 Adelie penguins distributed across the three islands. For analyzing the dataset, I used a couple of different methods:

Histograms, One-Way ANOVA, Two-Way ANOVA, ECDF, 3D plots, Scatter Plots, and Box Plots.

I used one-way ANOVA to statistically test whether the mean body measurements differed significantly across the three islands, with an alpha value of 0.05. This tests the null hypothesis that the population means are equal across all groups. I applied this same idea with a two-way ANOVA by adding sex as a second predictor. This helped me test the effects of island and sex on each body measurement.

In terms of data visualization techniques, I used bar blots to help compare average body measurements across islands and species. I also used scatter plots and box plots to help visualize the data points collectively to see trends in island groups and sex. Thirdly, I used ECDF plots to compare the distributions of each body measurement across islands. Finally, I used a 3D plot to compare each body measurement by island to see whether island groups form distinct clusters.

Data

1. Exploratory Analysis

Mean and SD for each penguin species

Species	Mean Bill Length (mm)	Mean Bill Depth (mm)	Mean Flipper Length (mm)	Mean Body Mass (g)	SD Flipper	SD Bill Length	SD Bill Depth
Adelie	38.82	18.35	190.1	3706.16	6.52	2.66	1.22
Chinstrap	48.83	18.42	195.82	3733.09	7.13	3.34	1.14
Gentoo	47.57	15	217.24	5092.44	6.59	3.11	0.99

Figure 1

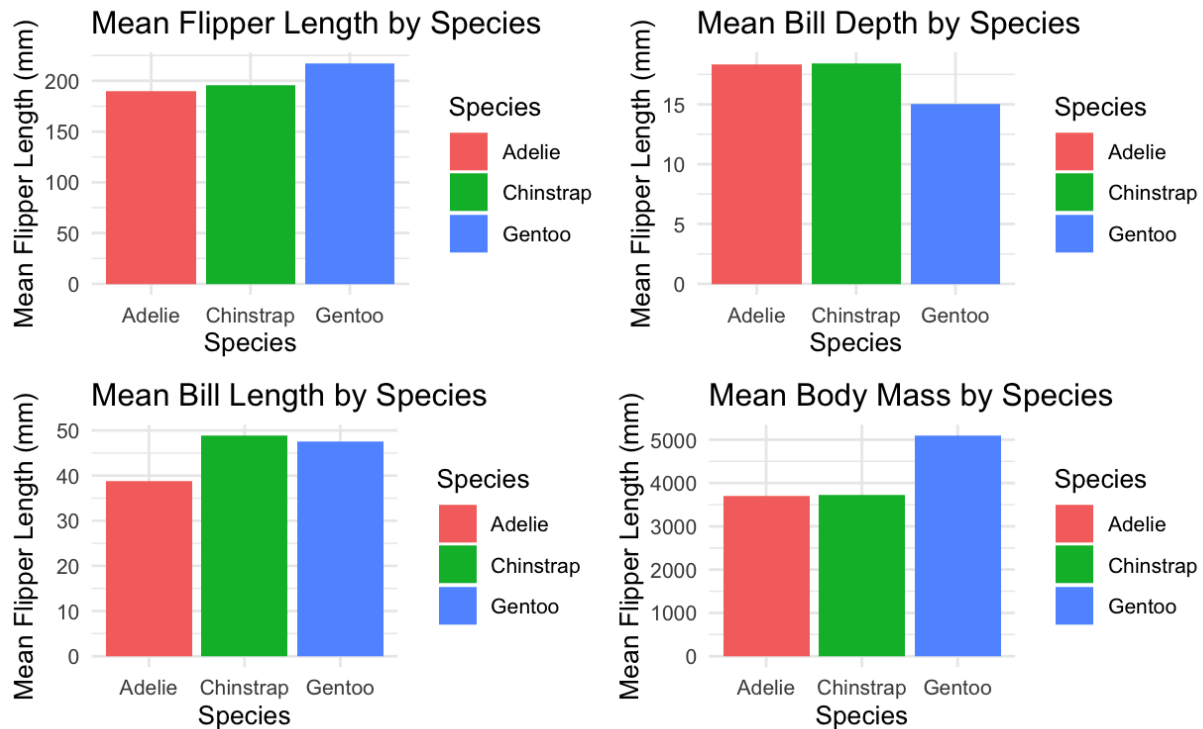


Figure 2

Before examining the Adelie penguin species independently from the other penguin species, I compared all three species together. The summary statistics and histogram of this are shown in Figures 1 and 2, displaying the mean body measurements for Adelie, Chinstrap, and Gentoo. Overall, I found that the Adelie and the Chinstrap species are very similar in average body measurements specially for Bill Depth, Body Mass, and Flipper Length. This suggests that they may exist in similar ecological environments. This is most likely true as the Adelie penguin lives with the Chinstrap penguin on the Dream Island.

Additionally, the Gentoo Penguins are larger on average, with the largest body mass and longest flippers. This may suggest that the Gentoo penguins might be able to swim faster and deeper, and may have different foraging and eating habits compared to the Adelie and chinstrap species.

The standard deviation values are also low among all species, showing that there is little variance in the data. The body measurements are tightly clustered within each species, showing strong consistency in measurements for each species.

2. Exploration of Question

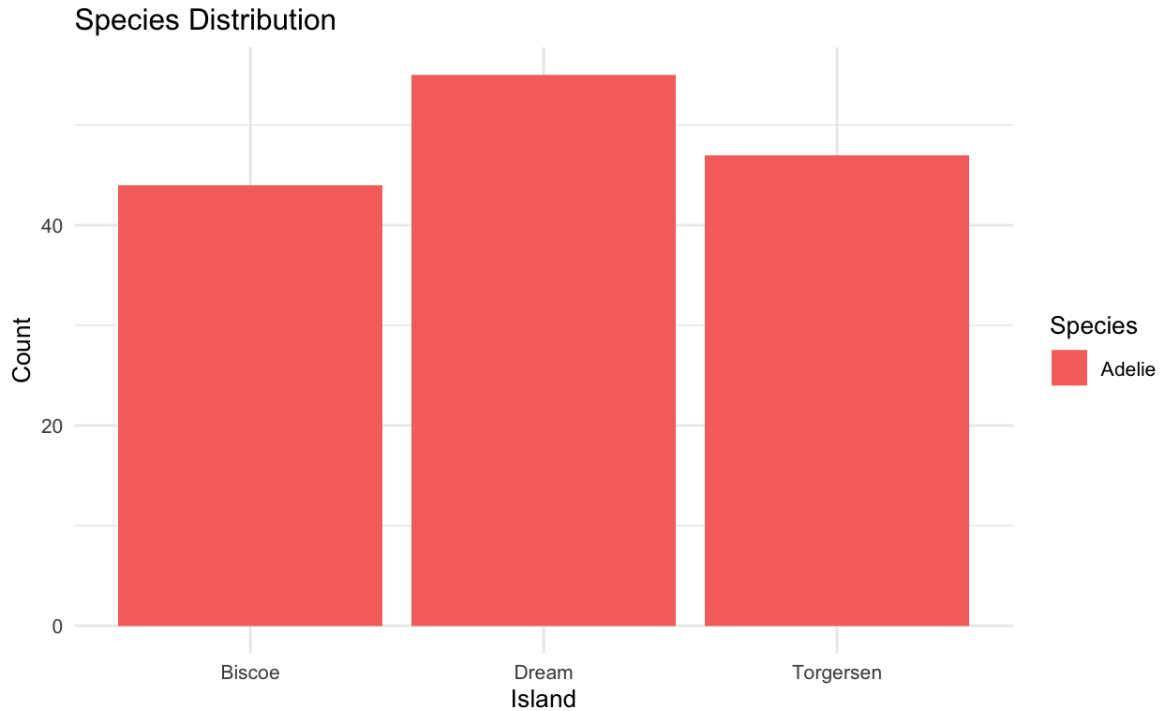


Figure 3

Mean and SD of Adelie measurements for each island

Island	Mean Bill Length (mm)	Mean Bill Depth (mm)	Mean Flipper Length (mm)	Mean Body Mass (g)	SD Flipper	SD Bill Length	SD Bill Depth
Biscoe	38.975	18.3704	188.7955	3709.659	6.7292	2.4809	1.1888
Dream	38.52	18.24	189.9273	3701.364	6.4803	2.4843	1.1406
Torgersen	39.0383	18.4511	191.5319	3708.511	6.2201	3.0281	1.3465

Figure 4

Figures 3, 4, 5, 6, and 8 represent my exploration of my research hypothesis. Figure 3 shows the distribution of Adelie penguins across the three islands, with Dream being the most heavily populated with Adelie Penguins. This is intriguing in that since Torgersen is the only island with Adelie penguins, it would have a larger population of them since they face less competition with other penguin species, but this is not the case.

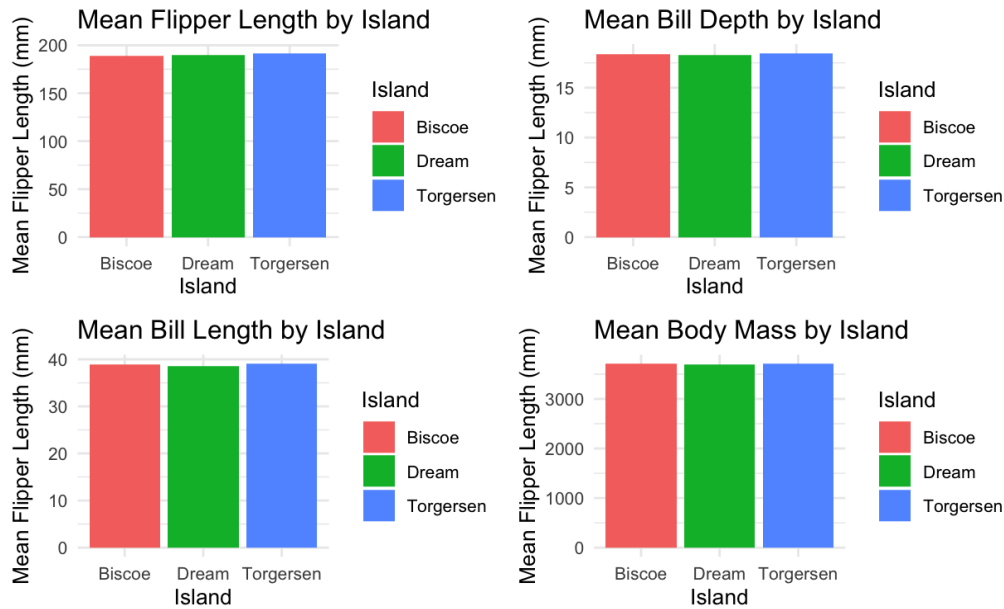


Figure 5

Figure 4 and Figure 5 complement each other and both demonstrate the mean body measurements of Adelie penguins on different islands. The average values for each body measurement are very similar across each island. The mean body mass is in a range of 10 grams for each island, and for mean flipper length, there is about a 1 mm difference in length across islands. The same idea is also applied to the mean bill length and depth.

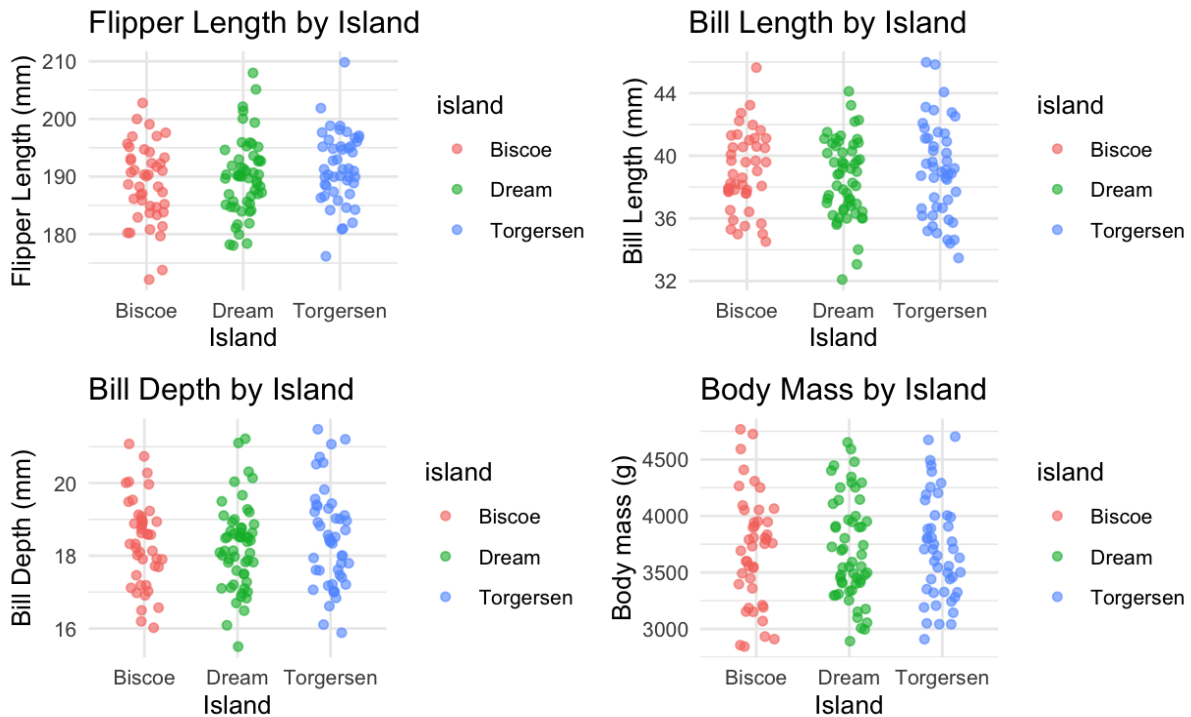


Figure 6

Figure 6 shows a scatter plot of the individual measurements by island. The data points show points from each island occupying the same area on each chart. If the island location affected body sizes, each island would have occupied different spaces on the graph.

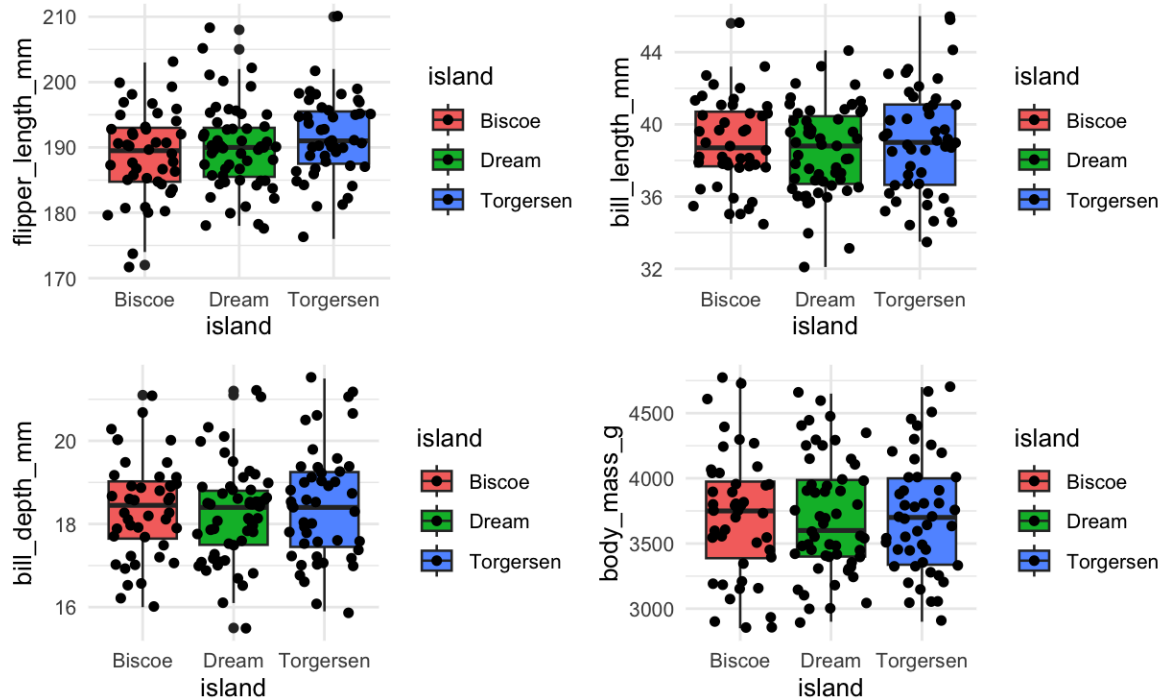


Figure 7

Figure 7 is similar and shows the distribution of the data. It demonstrates that the median body mass for Adelie is lowest on Dream, but other metrics are very similar on other islands. The variance in the median is not significant and can be accounted for by random sampling variation. This will also be confirmed by One-Way ANOVA. Additionally, the whisker and interquartile range interlap on each island, further showing no difference in body measurement for Adelie penguins.

Adelie Penguins: Bill Length vs Bill Depth vs Flipper Length

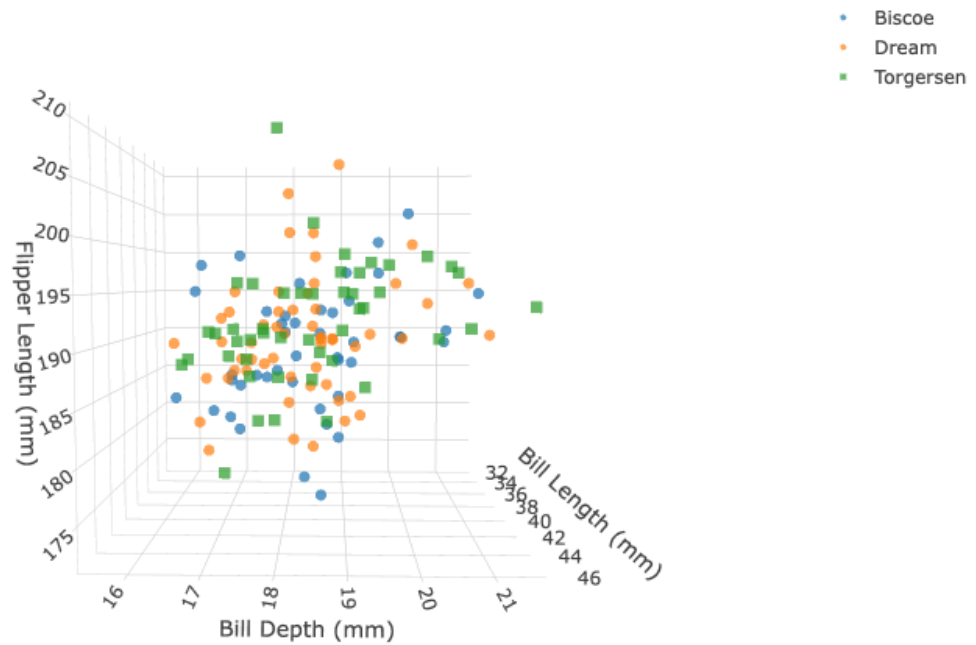


Figure 8

The 3D scatter plot displays the three islands, Biscoe, Dream, and Torgersen, and the four body measurements. There is no clustering by island, as all islands are intermixed in the 3D space. This again shows that there is no difference in body measurements by island location. This view of the data confirms across all factors at once that there is no significant effect of island location on Adelie penguin body size.

3. Statistical Testing on the Island Effect

One-way ANOVA Analysis based on body measurements and island location

Response Variable	Df	Sum Sq	Mean Sq	F value	Pr(>F)
flipper_length_mm ~ island	2	173	86.44	2.062	0.131
bill_length_mm ~ island	2	8.2	4.122	0.578	0.562
bill_depth_mm ~ island	2	1.16	0.5814	0.388	0.679
body_mass_g ~ island	2	2064	1032	0.005	0.995

Figure 9

Figure 9 shows the one-way ANOVA table, testing whether island predicts each body measurement. The table shows high p-values for every body measurement, as p-values are > 0.05 in the table. This leads us to fail to reject the null hypothesis, meaning there is no statistical difference in body measurement across the three islands. Additionally, all F-Values are very small, showing that islands play little role in explaining variance in body sizes.

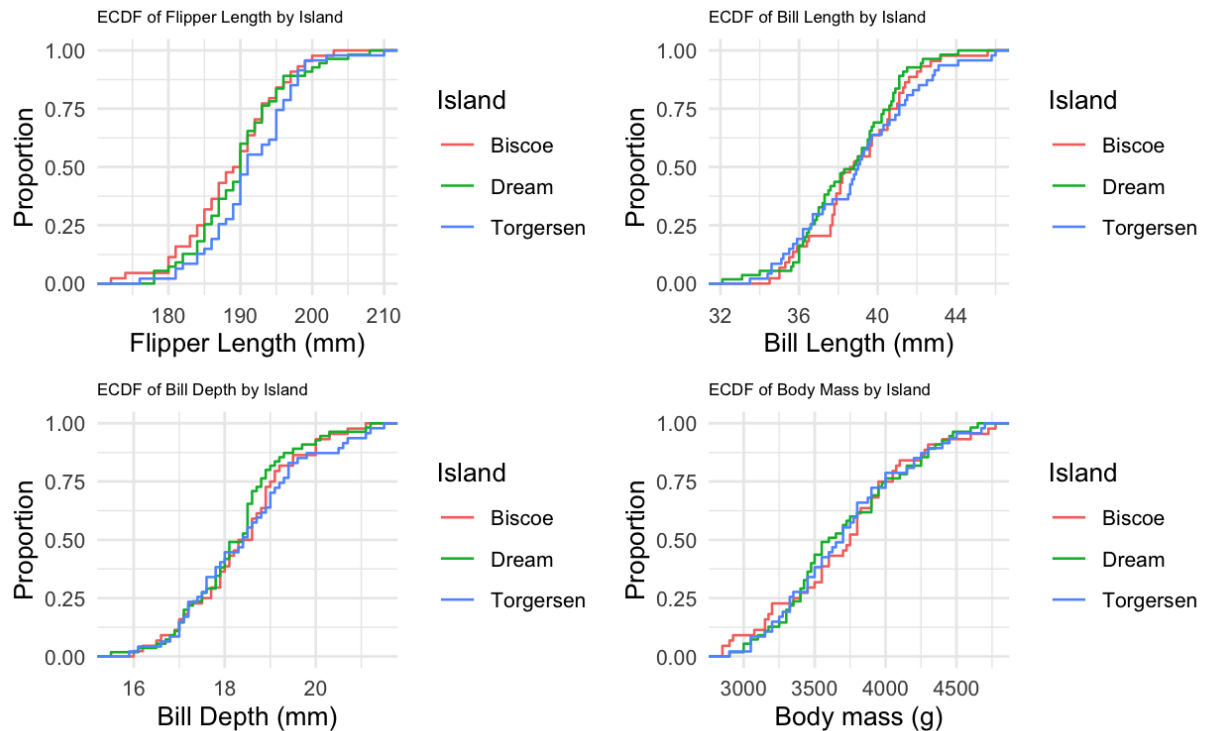


Figure 10

Figure 10 helps to further reinforce these ideas by visualizing the result. In the ECDF plots, all three lines overlap tightly across the whole graph. This shows the full distribution is similar across each island, and not only similar in the mean.

4. Exploration of sex and the island

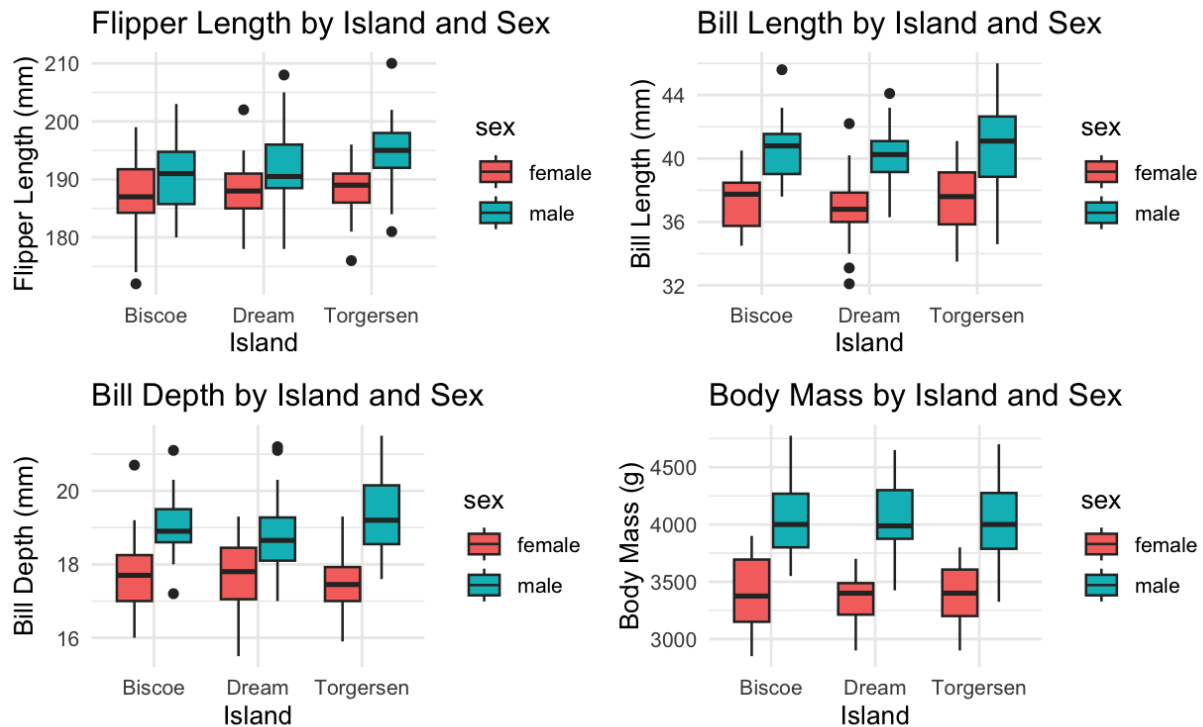


Figure 12

Figure 12 displays a box plot that shows the difference in body length measurements between male and female Adelie penguins. There is a clear division between male and female penguins for each body length, with all male penguins having larger body measurements in every measurement. An important note is that each median line up with each other by sex for each island, again showing that there is no significant influence by island on body measurements.

5. Significance of sex and island on body measurements

Two-way ANOVA examining the island and Sex and its effect on body measurements

Response Variable	Term	Df	Sum Sq	Mean Sq	F value	Pr(>F)
body_mass_g	island	2	2064	1032	0.011	0.989
	sex	1	16622756	16622756	170.141	<2e-16 ***
bill_depth_m	island	2	1.16	0.58	0.601	0.549
	sex	1	77.14	77.14	79.793	1.96e-15 ***
bill_length_m	island	2	8.2	4.1	0.887	0.414
	sex	1	360	360	77.476	4.17e-15 ***
flipper_length_mm	island	2	173	86.4	2.356	0.0984.

	sex	1	786	785.5	21.413	8.27e-06 ***
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Figure 13

Figure 13 displays the two-way ANOVA table, which tests the effect of island and sex on each body measurement. The result shows that the island has no significant effect with high p-values > 0.05 , while sex has a significant effect with all p-values being below 0.05. There are also high f-values for sex on each body measurement, showing that sex accounts for variance within the species.

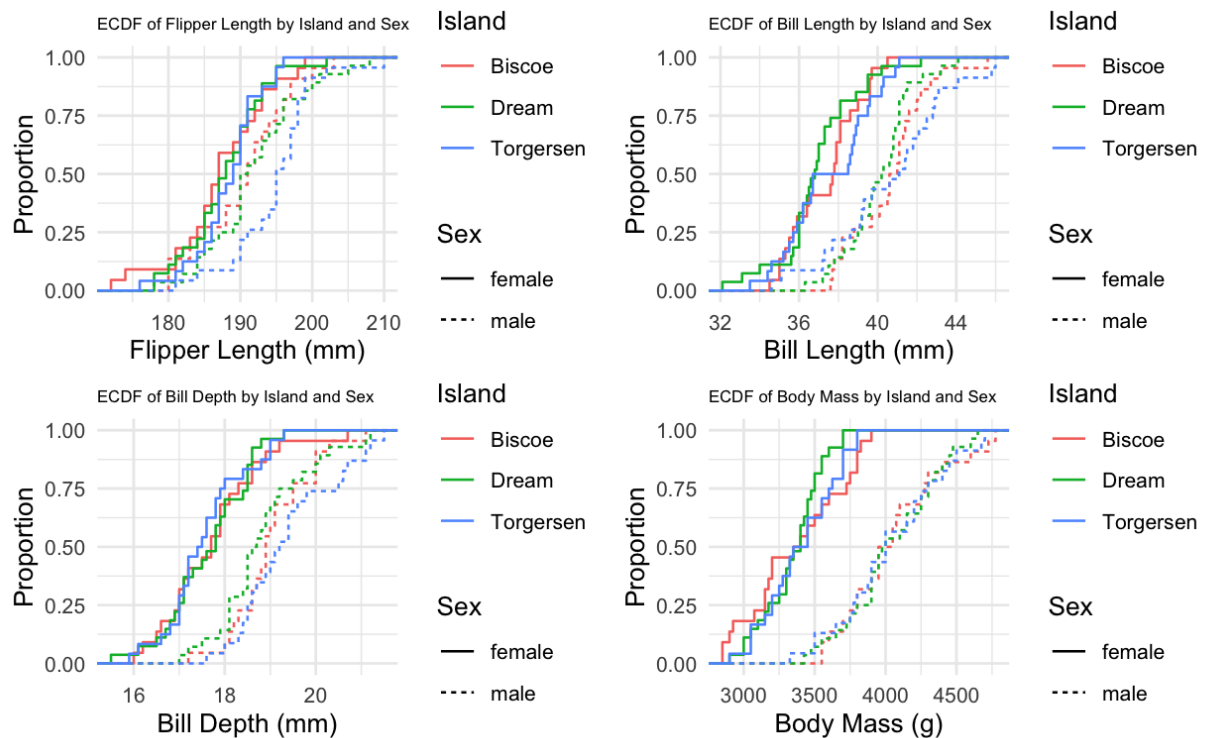


Figure 14

Figure 14 further reinforces the ideas shown in Figure 13. There is a clear divide in distribution between sexes in body measurements. The solid lines are female and clustered together, and the dotted lines are the males, also tightly clustered together. Between them is a large gap, showing the effect of sex on the distribution of body measurements. This shows the clear evidence of sexual dimorphism as a key reason for variance in the Adelie penguin species, and not island location.

Results

Overall, from the data exploration, I found that island location has no significant effect on Adelie penguin body measurements. Across all four body measurements and all three islands, distributions, means, and data points are nearly the same. This was shown with the one-way ANOVA test with all p-values lying above 0.05, as well as the ECDF plots, Histograms, and the 3D plot. These plots helped to give visual confirmation of the lack of effect of islands on body measurements.

Biologically, this could mean the food, climate, and environment on each island do not create morphological differences. Essentially, each island may have equivalent environments with similar food availability, predatory pressure, climate, and habitat structure. This means that each island did not create the selective pressures needed for the Adelie penguin to adapt to each island's environment. On the other hand, the morphological similarity of the Adelie penguins may explain why the species is the only one that can survive on each island. Perhaps it has the ideal body measurements that allows them to survive in more environments than other penguin species do. Another reason for the lack of morphological differences could be attributed to the Adelie penguin not being isolated on each island long enough. Local adaptation usually takes many generations and isolation to begin creating morphological differences.

In contrast, sex is shown to be a significant factor in Adelie body size. Male Adelie penguins are consistently larger than females across every island. This was proven with the two-way ANOVA table showing sex having p-values below 0.05. Additionally, the difference in body sizes by sex is consistent across each island, which further shows that the island has no significant effect on body size.

Conclusion

This analysis of the Adelie penguin found no support for my hypothesis that island location affects Adelie Penguin body measurements. All visualization techniques and ANOVA demonstrated that the island has no significant effect on body measurements.

The most meaningful result I found from this dataset was that sex is the primary driver for body size in Adelie penguins. Male Adelie penguins are consistently larger in all body measurements compared to females.

There are several limitations of my analysis. One is that I did not know the age of the penguins on each island or how long the colony has existed on each island. The age of the colony could

point to reasons as to why there are very little morphological differences on each island. The data also only spans three years, and this may not be enough time to observe biological change or evolution through the species. There is also no environmental data, such as predator density, temperature, harshness of climate, or food availability. There is no way for me to test if the islands are equivalent in terms of environment.

In the future, longer studies incorporating environmental data would help clarify whether islands have any influence on body measurements. Additionally, more frequent measurements and micro measurements in the body parts of the penguin may also help. Color and body width may also be factors to account for in future studies. GPS tagging on penguins could also be helpful to see if populations on islands are isolated or not.

Works Cited

**Aqua Expeditions. “A Short History of the Famous Darwin’s Finches.” *Aqua Expeditions*,
17 Apr. 2024, www.aquaexpeditions.com/blog/darwins-finches.**

“Torgersen Island.” *Polartours*, 2025, www.polartours.com/highlights/torgersen-island.